

Hypertension: If-Then Rules (LLM-generated draft)

Hypertension — if-then rules

Decision rules the chatbot can pattern-match against to give specific, non-generic guidance. Derived from the WHO *HEARTS technical package*, the WHO *Be He@lthy Be Mobile — mHypertension handbook* (2020), and WHO *treatment guidelines for hypertension*.

All numerical thresholds are simplified for adult patients without specific comorbidities. A clinician must individualise these for pregnancy, diabetes, chronic kidney disease, and elderly patients with orthostatic hypotension.

Triage by blood-pressure reading

If the most recent reading is ...	And the person ...	Then ...
< 120 / 80	(anyone)	Reassure: this is in the normal range. Encourage the lifestyle steps that maintain it.
120-139 / 80-89	not yet diagnosed	Flag as “elevated.” Recommend recheck within 1-3 months and the four lifestyle pillars (salt, fruit/veg, activity, no smoking, limit alcohol).
140-159 / 90-99	not yet diagnosed	Recommend they see a health worker within 1-2 weeks for confirmation. Do not start medication advice in the chat.
140-159 / 90-99	already diagnosed, on medication	Reinforce adherence. If consistently above 140/90 across multiple readings over 1-2 weeks, recommend a clinic visit to review medication.
160-179 / 100-109	(anyone)	Recommend a clinic visit within a few days. Reinforce adherence if on medication.
≥ 180 / 110	symptom-free	Recommend they sit quietly for 5 minutes and recheck. If still ≥ 180/110, contact health worker the same day .
≥ 180 / 110	with any red-flag symptom (see below)	Hypertensive emergency. Tell them to seek emergency care immediately — do not drive themselves.

Red-flag symptoms (hypertensive emergency)

If any of these appear with a high reading, escalate to emergency:

- Severe headache, especially sudden onset
- Loss of consciousness, confusion or sudden weakness
- Chest pain or pressure
- Nausea and vomiting that won't stop
- Sudden visual disturbance or loss of vision
- Racing or pounding heartbeat
- Sudden trouble speaking, slurred speech, drooping of one side of the face
- Severe shortness of breath
- Weakness or numbness on one side of the body (possible stroke)

Medication adherence

If the person reports ...	Then ...
missed yesterday's dose	"Take today's dose as usual. Do not double up. Set a reminder so it doesn't happen again."
missed several doses this week	"Take today's dose. Tell your health worker — your readings will be off and they may need to adjust. Ask about a reminder system or a combination pill."
stopping because "I feel fine"	Explain that hypertension is silent and the pills are why they feel fine. Strongly discourage stopping. Offer to set adherence reminders.
stopping because of side effects	Discourage stopping unilaterally. Ask which side effect. Suggest contacting the health worker — there is almost always an alternative within the same effect class.
can't afford the medication	Surface this as a barrier worth addressing. Recommend they ask their health worker about (a) generic equivalents, (b) lower-cost combination tablets, (c) any subsidy or NCD-programme access in their region. Do not advise stopping.
dry persistent cough on a "-pril" medicine (ACE inhibitor)	Explain this is a known side effect and an alternative class (ARBs, ending in "-sartan") exists. Recommend they ask the health worker to switch. Don't stop on their own.
swollen ankles on a "-pine" medicine (CCB)	Acknowledge this is a known side effect. Recommend the health worker review the dose or switch. Don't stop on their own.

Salt and diet

If the person says ...	Then suggest ...
they use stock cubes (Maggi, bouillon)	Cut to half a cube per pot, add later in cooking, replace flavour with onion, garlic, ginger, scotch bonnet, herbs, lemon.
they eat bread every morning	Note bread is a hidden salt source. Suggest 3-4 mornings a week instead of 7, and pair with avocado/egg instead of butter and processed meat.
"I don't add salt at the table"	Acknowledge but redirect: most dietary salt comes from cooking and processed foods, not the salt shaker. Focus on stock cubes, processed meat, dried/salted fish, instant noodles, canned foods.

If the person says ...	Then suggest ...
their meals are mostly rice, oil and a bit of meat	Recommend adding vegetables (any colour, raw or cooked) to half the plate, swapping palm/coconut oil for sunflower/soya/olive, and using fish or chicken more often than red meat.
they drink sugary drinks daily	Recommend water as the default, fresh fruit instead of juice, and reducing sugar in tea/coffee gradually.

Portion guidance (5-a-day)

- 1 portion = 1 banana, 1 mango, 1 apple, 1 orange, 1 handful of berries, 3 tablespoons of cooked vegetables, or a cup of leafy greens.
- **Starchy roots** (potato, sweet potato, cassava, yam, plantain) do **not** count toward the 5.
- Juice counts at most as 1 portion per day, regardless of how much is drunk.

Physical activity

If the person ...	Then recommend ...
has done no exercise in years	Start with 10-minute walks, twice a day. Add a minute or two per week. Goal: 30 min/day, 5 days/week.
does some activity but less than 150 min/week	“Add one or two extra 15-minute walks. Try to make it brisk enough that you breathe faster.”
already meets 150 min/week	Encourage maintenance and variety (walking, gardening, dancing, swimming, cycling).
has knee/joint pain	Suggest swimming, cycling, or shorter more frequent walks on flat ground with cushioned shoes. Recommend telling the health worker if pain persists.
has chest pain on exertion, severe shortness of breath, or dizziness when exercising	Stop and see a health worker before continuing. Do not push through these symptoms.
current reading > 180/110	Defer exercise advice. Recommend they get their pressure under control first via clinic visit.

Tobacco, alcohol, weight

If the person ...	Then ...
smokes	Strongly recommend quitting as the single highest-impact change. Offer to refer them to the tobacco-cessation flow. Do not minimise.
has someone in the home who smokes	Recommend asking them not to smoke indoors or in the car. Second-hand smoke is a real cardiovascular risk for them too.
drinks alcohol daily	Recommend at least 2 alcohol-free days per week, and no more than 2 drinks on any drinking day. The lowest-risk option remains zero.
drinks > 14 drinks/week or has trouble cutting back	Treat as a flag. Encourage them to discuss with their health worker; there is help available beyond willpower.

If the person ...	Then ...
is overweight	Recommend slow, steady weight loss (about 0.5 kg / 1 lb per week) through eating and movement combined. Discourage crash diets and miracle products.
is underweight	Note: this is not the typical hypertension picture and may indicate something else (uncontrolled diabetes, infection, malnutrition). Recommend a clinic visit.

Self-monitoring at home

If the person ...	Then ...
has a home BP machine and asks how to use it	Sit quietly for 5 min, feet flat on the floor, arm at heart level, cuff on bare skin, don't talk during the reading. Take 2 readings 1 min apart and record both. Best times: morning before medication, and evening.
reports very different readings on the same arm	Reassure that BP varies through the day. Recommend the same time, same posture, same arm each day; record the average over a week rather than reacting to single readings.
asks how often to check at home	If newly started on treatment: daily for the first 1-2 weeks, then 2-3 times/week once stable. If stable for months: 1-2 times/week is usually enough. Always more often if symptoms appear.
asks "is this reading bad?"	Use the triage table above. Do not interpret a single reading in isolation — encourage them to take 3-5 readings over a few days before drawing conclusions.

Things to never do in the conversation

- Never recommend stopping prescribed medication.
- Never recommend a specific new prescription medication by name.
- Never provide dose changes for blood pressure medicines — that is a clinical decision.
- Never claim hypertension can be "cured" by a particular food, herb, supplement, or programme.
- Never minimise red-flag symptoms or delay the emergency-care recommendation when the triage rules above call for it.
- Never give clinical advice that contradicts what the patient's own health worker has told them — surface the conflict and recommend they clarify with the health worker.